

Q1 - 28 June - Shift 1

Let AB and PQ be two vertical poles, 160 m apart from each other. Let C be the middle point of B and Q, which are feet of these two poles. Let $\frac{\pi}{8}$ and θ be the angles of elevation from C to P and A, respectively. If the height of pole PQ is twice the height of pole AB, then $\tan^2 \theta$ is equal to

- (A) $\frac{3-2\sqrt{2}}{2}$ (B) $\frac{3+\sqrt{2}}{2}$
(C) $\frac{3-2\sqrt{2}}{4}$ (D) $\frac{3-\sqrt{2}}{4}$

Space for your notes:

Q2 - 29 June - Shift 2

From the base of a pole of height 20 meter, the angle of elevation of the top of a tower is 60° . The pole subtends an angle 30° at the top of the tower. Then the height of the tower is:

- (A) $15\sqrt{3}$ (B) $20\sqrt{3}$
(C) $20+10\sqrt{3}$ (D) 30

Space for your notes:

Questions

MathonGo

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Answer Key

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Q1 (C)

Q2 (D)

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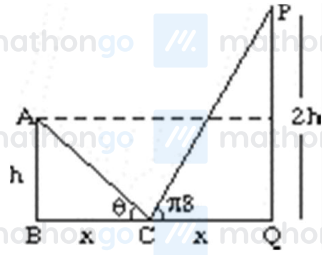
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Q1 (C)



Let $BC = CQ = x$ & $AB = h$ and $PQ = 2h$

$$\tan \theta = \frac{h}{x}, \tan \frac{\pi}{8} = \frac{2h}{x}$$

$$\frac{\tan \theta}{\tan \left(\frac{\pi}{8} \right)} = \frac{1}{2}$$

$$\tan \theta = \frac{1}{2} \tan \left(\frac{\pi}{8} \right) = \frac{1}{2} (\sqrt{2} - 1)$$

$$\tan^2 \theta = \frac{1}{4} (3 - 2\sqrt{2})$$

Q2 (D)

$$PT = \frac{h}{\sqrt{3}} = AB$$

$$\frac{AB}{h - 20} = \sqrt{3}$$

$$h = 3(h - 20)$$

$$h = 30$$

